Association Rules

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HSE

Course Objectives

- **What Are Association Rules?**

  Find items or attributes that occur together
- **How Do We Evaluate a Rule?**

  Calculate support · confidence · lift
- **Why Does Lift Matter?**

  Compare a rule’s confidence with the baseline rate
- **How Should We Interpret Results?**

  Distinguish association from causation

If They Buy Diapers, What Else?

Discovering patterns in shopping baskets



Diapers → Beer?

Do these items really appear together more often than expected?

Start With One Simple Basket

Before formulas: what exactly counts as a transaction?

8 transactions

Basket	Items
T1	A, B
T2	A, B
T3	A, B
T4	A
T5	B
T6	B
T7	A
T8	A, B

For A → B

Count only baskets where A and B occur together.

A and B = 4 baskets

T1 • T2 • T3 • T8

Same customer ≠ same transaction.
A rule uses the basket as the unit.

One Row = One Transaction

Analyze products purchased together in a single basket.

Transaction	Customer & day	Items in this basket
T1	Alex · Mon	Diapers , Beer
T2	Sam · Mon	Bread , Milk
T3	Alex · Fri	Diapers , Wipes
T4	Lee · Fri	Beer , Chips
T5	Sam · Sat	Bread , Eggs
T6	Lee · Sat	Diapers , Milk

Same customer, different baskets

Count items together only when they occur in the same transaction.
Do not combine purchases from different days into a single basket.

Three Questions About a Rule

$$A \rightarrow B$$

01

SUPPORT

How often do A and B appear together?

$P(A \text{ and } B)$

Out of all transactions



02

CONFIDENCE

When A appears, how often does B appear too?

$P(B \mid A)$

Out of transactions containing A



03

LIFT

Is B more common with A than B is overall?

$P(B \mid A) / P(B)$

Compare with B's baseline frequency



Calculate an Association Rule: A → B

Four shopping transactions

Basket	Items
T1	A, B
T2	A, B
T3	A only
T4	Neither A nor B

All = 4 A = 3 B = 2 A and B = 2

SUPPORT

$$P(A \text{ and } B) = \frac{2}{4} = 0.50$$

2 baskets contain both A and B, out of 4 total.

CONFIDENCE

$$P(B \mid A) = \frac{2}{3} \approx 0.67$$

2 baskets contain B, out of 3 baskets with A.

LIFT

$$\frac{P(B \mid A)}{P(B)} = \frac{(2/3)}{(2/4)} = \frac{4}{3} \approx 1.33$$

B occurs in 2 of 4 baskets overall.

Lift > 1: B is more frequent among baskets with A (67%) than among all baskets (50%).

Yet Another Example → Three Numbers

Use the same 8 baskets for support, confidence, and lift.

A and B	A	B	All baskets
3	5	5	8

SUPPORT — **3 / 8 = 37.5%** — How often A and B appear together

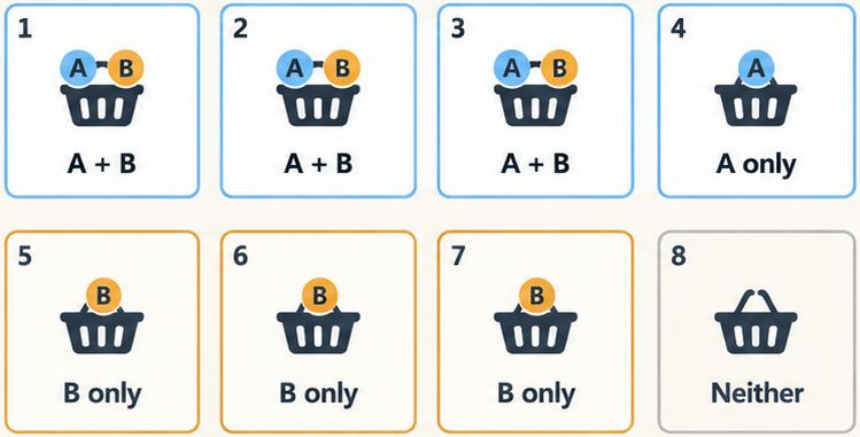
CONFIDENCE — **3 / 5 = 60%** — Among A baskets, how often is B present?

LIFT — **60% / 62.5% = 0.96** — Compare confidence with B's baseline

Key point: 60% confidence sounds reasonable, but B already appears in 62.5% of all baskets → lift < 1.

Why Confidence Alone Can Mislead

A 75% confidence sounds high—until you check the baseline.



WITH A

3 of 4 = 75%

Confidence: B among A baskets

OVERALL

6 of 8 = 75%

Baseline: B among all baskets

$$\text{Lift} = \frac{75\%}{75\%} = 1.00$$

A does not make B more likely than B already is.

Illustrative transactions; association is not causation.

From Baskets to Candidate Rules

1 Transactions

T	Basket
T1	A, B, C
T2	A, B, C
T3	A, B
T4	A, C

N = 4 baskets

2 Find an itemset

{A, B, C}

Appears in T1 and T2

Support = 2 / 4 = 50%

An itemset is unordered: no arrow yet.

3 Split into rules

{A, B} → C

{A, C} → B

{B, C} → A

Same itemset, different directions and confidence.

First find items that appear together; then choose a left side and a right side to create candidate rules.

These are candidates, not yet accepted recommendations: evaluate confidence and lift.

A → B Is Not the Same as B → A

Support is symmetric; confidence is directional.

A → B

3 baskets contain A and B

5 baskets contain A

Confidence = 3 / 5 = 60%

A is the condition.

B → A

3 baskets contain A and B

6 baskets contain B

Confidence = 3 / 6 = 50%

B is the condition.

If A appears in 5 baskets but B appears in 6, the denominators change → A → B = 60%, while B → A = 50%.

So What Does Lift Tell Us?

Always ask: “Compared with what?”

Suppose 90% of A-buyers also buy B.

Confidence = 90%

But B is already very common

B appears in 80% of ALL baskets

Baseline P(B) = 80%

So what does A add?

Lift = 90% / 80% = 1.125

Only a small increase over the baseline.

Confidence asks “How often B follows A?” • Lift asks “Is B unusually common when A occurs?”

Thresholds: More Rules ≠ Better Rules

Minimum support and confidence control how many candidates survive.

MINIMUM SUPPORT

Higher support

→ fewer rules
→ more evidence per rule

Lower support

→ more rare patterns
→ more noise / computation

MINIMUM CONFIDENCE

Higher confidence

→ rules are more predictable
→ but may miss useful rules

Lower confidence

→ more rules survive
→ many may be weak

There is no universal threshold. State your threshold and justify the trade-off.

Rule Selection: Three Gates

Toy data, 4 baskets: T1 {A,B,C} T2 {A,B,C} T3 {A,B} T4 {A,C}
Example rule: B → A

1 SUPPORT

Are B and A together often enough?

support(A,B) = 3/4 = 75%

min support ≥ 50%

✅ **PASS**

2 CONFIDENCE

When B occurs, how often does A occur?

confidence(B → A) = 3/3 = 100%

min confidence ≥ 75%

✅ **PASS**

3 LIFT

Is A more likely than its baseline?

lift(B → A) = 100% / 100% = 1.00

require lift > 1

❌ **FAIL**

Rule Check (B → A)

Support: 3/4 = 75% ✅ (≥ 50%) → Confidence: 3/3 = 100% ✅ (≥ 75%) → Lift: 1.00 ❌ (> 1)

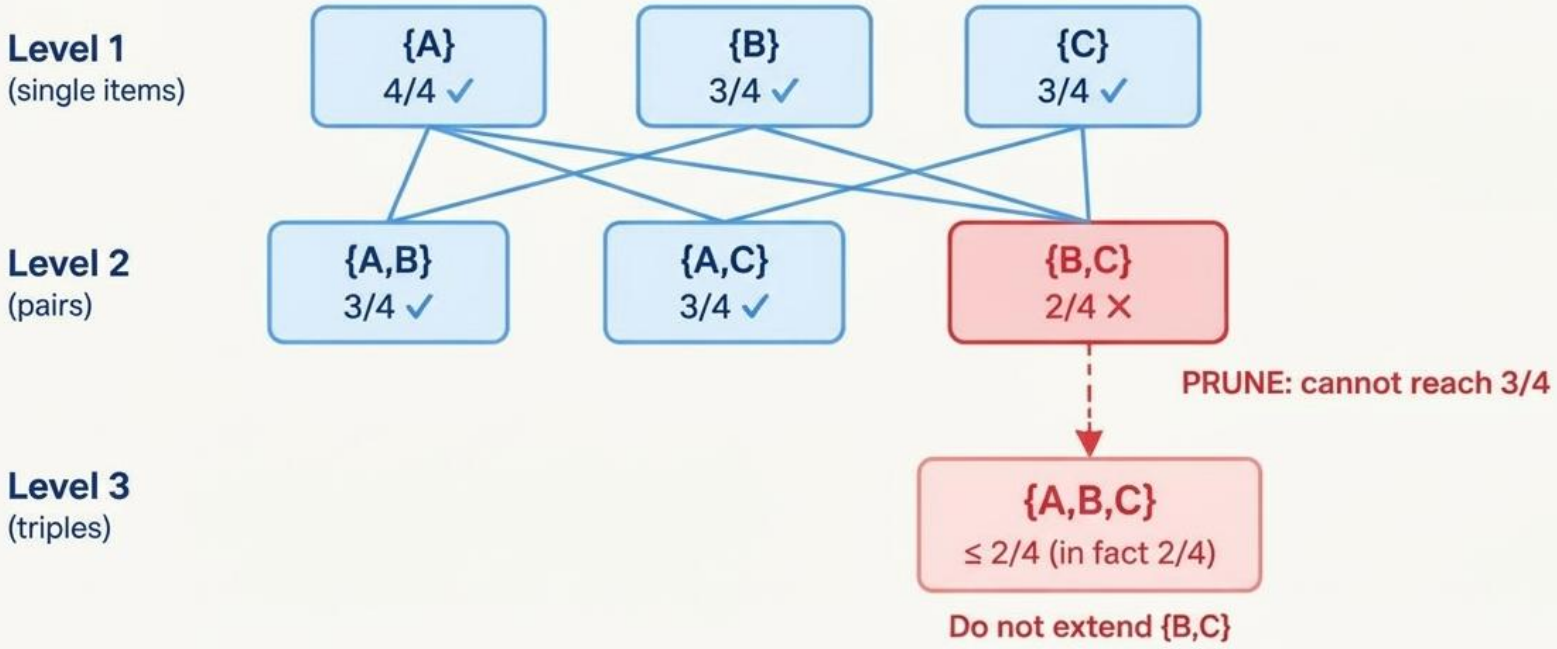
💡 **High confidence can be misleading: A appears in every basket, so B adds no information.**

Rule passes support and confidence but fails the positive-lift filter.

Apriori: Prune Before You Search

Transactions (N = 4): T1 {A,B,C} T2 {A,B,C} T3 {A,B} T4 {A,C}

Minimum support = 3 of 4 baskets



If a set is infrequent, every larger set containing it is also infrequent.

Stop expanding {B,C}; skip counting its supersets as frequent candidates.

FP-Growth: Mine Patterns Without Candidate Generation

Transactions (N = 5): T1 {A,B,C} | T2 {A,B} | T3 {A,B,C} | T4 {A,C} | T5 {B,C}

Minimum support = 3 / 5

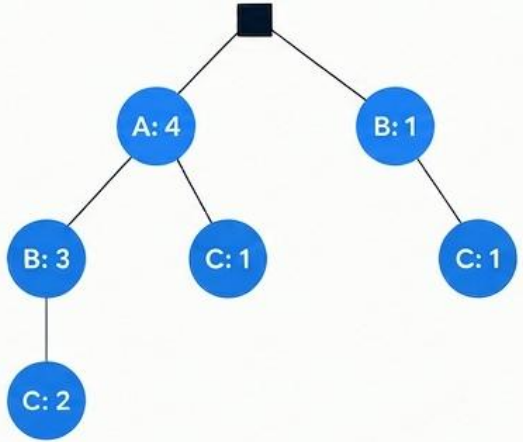
1. COUNT + ORDER

A: 4 B: 4 C: 4

Tie-break order: A, then B, then C

Sort items in each basket before inserting.

2. BUILD ONE FP-TREE



Shared prefixes share one path.

3. MINE FREQUENT SETS

Itemset	Conditional Pattern Base (CPB)
C:	(A,B:2), (A:1), (B:1)
B:	(A:3)
A:	(ø:4)
Final Frequent Sets (Supp >= 3)	

✔ {A} = 4
✔ {B} = 4
✔ {C} = 4
✔ {A,B} = 3
✔ {A,C} = 3
✔ {B,C} = 3
✘ {A,B,C} = 2

Only sets with count at least 3 survive.

FP-Growth stores repeated prefixes once, then mines frequent itemsets from the FP-tree.

Association rules are generated after frequent itemsets are found.

Validation Before Action

Past data | Later data | Experiment

1 DISCOVER

Mine rules from past transactions

support · confidence · lift

2 VALIDATE

Check rules on a later time period

Do support and lift hold up?

3 TEST

Run a controlled cross-sell or bundle test

Compare outcomes to a control group

4 ACT

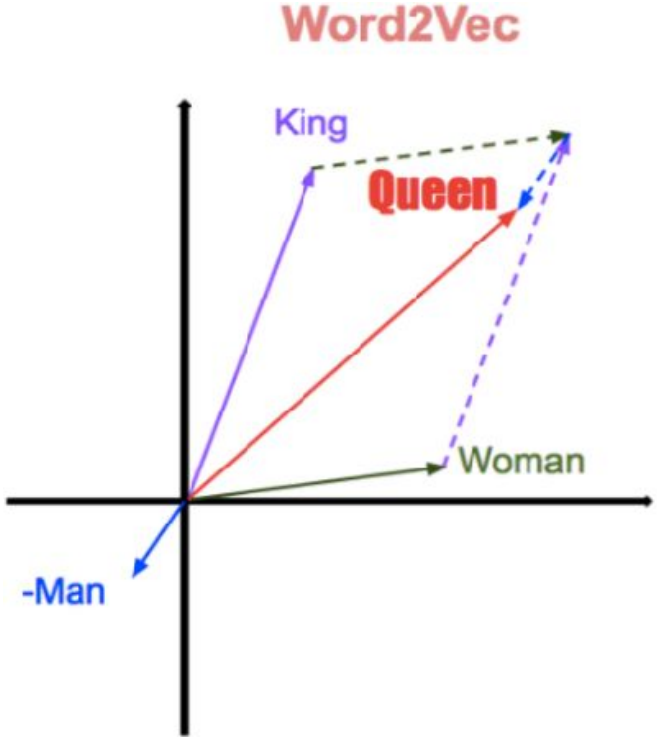
Deploy only if incremental value is positive

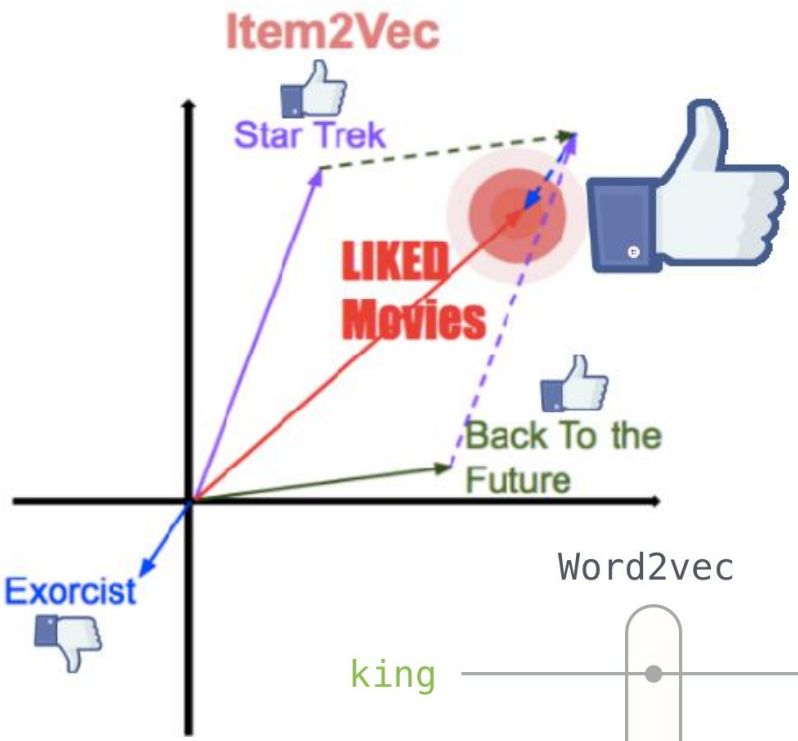
Consider margin, stock, and customer experience

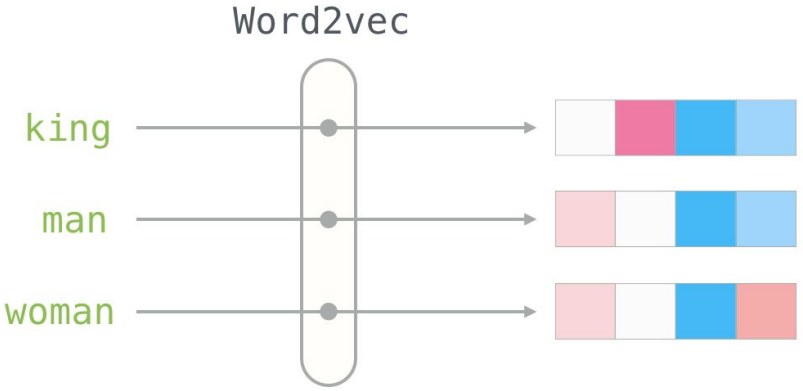
Co-occurrence is not causation. A high lift in past data does not prove an intervention will increase sales.

Extra

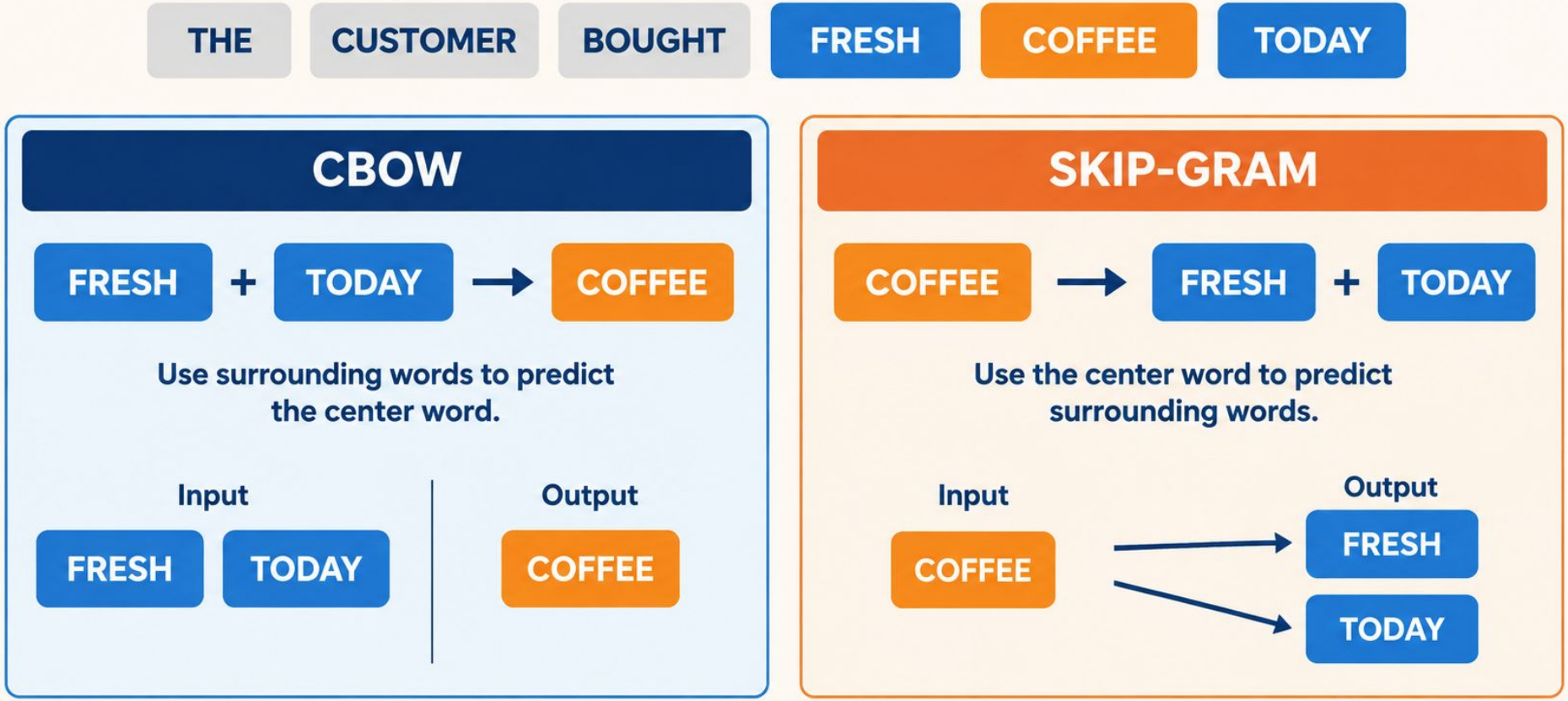
Co-Occurrence and Word2Vec (Item2Vec)







Co-occurrence: Words or items that appear in similar contexts provide signals for learning.
Word2Vec: Learns a numerical vector for each word from the words around it.
Vector relationships: Similar contexts can produce meaningful patterns in the embedding space; the king–queen diagram is a conceptual illustration.



[1301.3781] Efficient Estimation of Word Representations in Vector Space

Item2Vec: Learning Similar Movies

Skip-gram learns item vectors from movies viewed in the same user context.

1 Context from user sessions

S1: Star Trek · Back to the Future

S2: Star Trek · Back to the Future

Star Trek ⟶ Back to the Future

Observed positive pair

Star Trek - - - - - - - - - - ⟶ Exorcist

Sampled noise pair

Noise does not mean dislike.

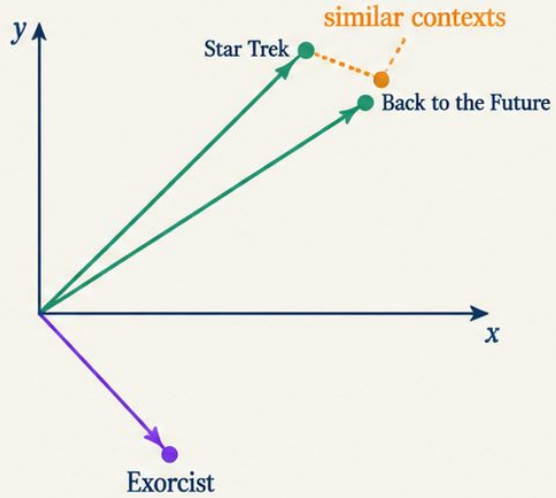
2 Skip-gram + negative sampling

$$\text{maximize } \log \sigma(v_{\text{ST}} \cdot u_{\text{BTTF}}) + \log \sigma(-v_{\text{ST}} \cdot u_{\text{Exorcist}})$$

Increase score for the observed context item.

Decrease score for a sampled noise item.

3 Learned movie vectors



Find neighbors using cosine similarity.

Conceptual illustration: real vectors require many sessions; these positions are not computed from the toy example.

Negative Sampling: Learn from One Match and a Few Contrasts

1. Build a training question

Observed session:
Star Trek + Back to the Future

Pair	Target
(Star Trek, Back to the Future)	1: observed
(Star Trek, Exorcist) sampled from catalog, not observed together in THIS session.	0: sampled

⚠ 0 means sampled contrast, NOT a dislike or a true negative.

2. Update the scores

$$p(\text{context} \mid \text{target}) \approx \sigma(v_{\text{target}} \cdot u_{\text{context}})$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\text{maximize} \quad \log \sigma(v_{ST} \cdot u_{BTTF}) + \log \sigma(-v_{ST} \cdot u_{Exorcist})$$

v: target embedding; u: context embedding

\+ increase observed-pair score

− decrease sampled-pair score

Note: This is for ONE positive and ONE sampled negative only, not a full multi-negative objective.

3. Why sample?

Full softmax:
score EVERY catalog item for EACH observed pair

$O(|\text{catalog}|)$
scores per example

Negative sampling:
score the observed item + k sampled items

$O(k + 1)$
scores per example

If catalog = 10,000 and k = 5: evaluate 6 scores instead of 10,000 per training example *(illustrative)*.

After many updates: similar viewing contexts can yield useful movie embeddings; sampled negative ≠ disliked movie.

Association Rules vs. Item2Vec

Both use co-occurrence signals — but they learn different things.

ASSOCIATION RULES		ITEM2VEC	
Input	Transactions / baskets	**Input**	Sequences / co-occurring items
Output	Explicit rules: A → B	**Output**	Dense vector for each item
Question	Are A and B unusually associated?	**Question**	Which items have similar contexts?
Evaluation	support • confidence • lift	**Evaluation**	Similarity / downstream ranking
Typical use	Cross-sell / bundle / pattern discovery	**Typical use**	Recommendation / retrieval

Same idea: co-occurrence provides a signal. Different goal: rules explain relationships; embeddings represent items.

Same behavior data, different outputs

[1603.04259] Item2Vec: Neural Item Embedding for Collaborative Filtering

Repeated user / session histories (co-occurrence data)

S1: Star Trek – Back to the Future
S2: Star Trek – Back to the Future
S3: Exorcist

Items that appear in the same session co-occur. This information will be used in different ways by the two methods.

ASSOCIATION RULES

Co-occurrence → Explicit rules

1. Find co-occurrence from sessions → 2. Mine association rules → 3. Evaluate with support, confidence, lift

Co-occurrence pairs

S1: Star Trek + Back to the Future
S2: Star Trek + Back to the Future
S3: Exorcist

→

Learned rule (example)

Star Trek → Back to the Future

Support	2/3 = 66.7%
Confidence	2/2 = 100%
Lift	1.5

? ***Question:*** *How often does B occur when A occurs?*

ITEM2VEC

Co-occurrence → Learned embeddings

1. Use sessions to create co-occurrence context → 2. Train item embeddings (e.g., predict context items) → 3. Items with similar contexts get similar vectors

Shared context (from sessions)

S1: Star Trek ↔ Back to the Future
S2: Star Trek ↔ Back to the Future
S3: Exorcist

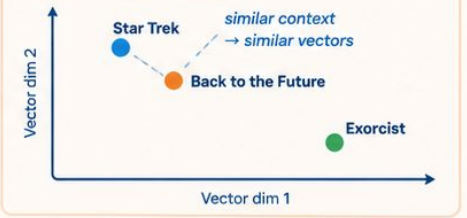
↓

Training pairs (co-occurrence)

(Star Trek, Back to the Future), ...

→

Learned item embeddings



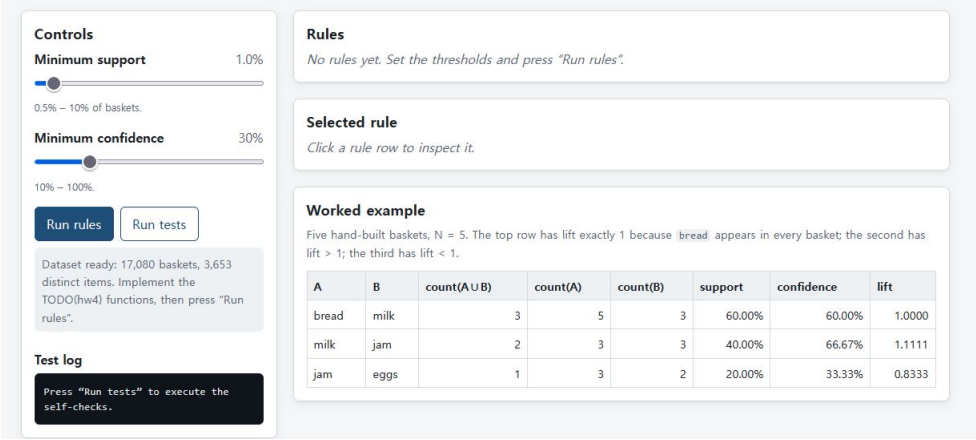
? ***Question:*** *Which items have similar viewing contexts?*

Both methods use co-occurrence data, but in different ways:

Association Rules = co-occurrence → **explicit rule** (A → B)
Item2Vec = co-occurrence → learned representation (A ≈ B)

Discussion

HW4. Association Rules



1. Implementation Task

- **Repository:** [dryjins/RecSys-LLMs/tree/main/week4]
- **Dataset:** [UCI Online Retail dataset]. Treat each InvoiceNo as one transaction; use StockCode to identify items and Description to label them.
- **Basket Construction:** Create an invoice–item basket matrix.
- **Association Rule Mining:** Use Apriori or FP-Growth to identify frequent itemsets and generate association rules.
- **Rule Evaluation:** Report support, confidence, and lift for the selected rules.
- **Rule Filtering:** State and justify your minimum support and confidence thresholds; retain rules with **lift > 1** for further analysis.

2. Business & Algorithmic Analysis

- **(Mandatory) Useful vs. Misleading Rules:** Present one rule worth investigating and one rule to reject. Justify both using transaction counts, support, confidence, and lift.
- **(Below Optional)**
- **Rule Direction:** Compare **A** → **B** with **B** → **A**. Explain why their confidence values can differ.
- **Threshold Trade-off:** Explain how changing minimum support or confidence affects the number and usefulness of discovered rules.
- **Validation Before Action:** Check whether a selected rule remains stable in a later time period. Propose a controlled cross-sell, bundle, or product-placement experiment.
- **Association ≠ Causation:** Explain why high lift alone does not prove that an intervention will increase sales.

HW3. Collaborative Filtering

1. Implementation Task

- **Repository:** dryjins/RecSys-LLMs/tree/main/week3
- **Dataset:** Same MovieLens 100k files reused from Week 2 (u.data, u.item — byte-identical)
- **User-Based & Item-Based CF:** Implement similarity calculation comparing users (columns) vs items (rows)
- **Missing Value Handling:** Choose a strategy (mean imputation, weighted by common ratings, or matrix factorization) for sparse ratings

2. Business & Algorithmic Analysis

- **User-based vs. Item-based:** Compare recommendation quality and efficiency depending on whether number of users or items is larger
- **Missing-Value Strategy Trade-off:** Analyze simplicity vs. bias vs. computing cost across mean imputation, weighted approach, and matrix factorization
- **Cold Start and Sparsity:** Explain why CF fails for new users or items with no ratings, and why similarity scores computed from very few co-rated items are unreliable (too little evidence).

